Pontificia Universidad
JAVERIANA
Bogotá

| VIGILADA MINEDUCACIÓN |

# Instrumentación para la agroindustria


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Facultad de ingeniería

# Introduction

¿Why technologies in agriculture?

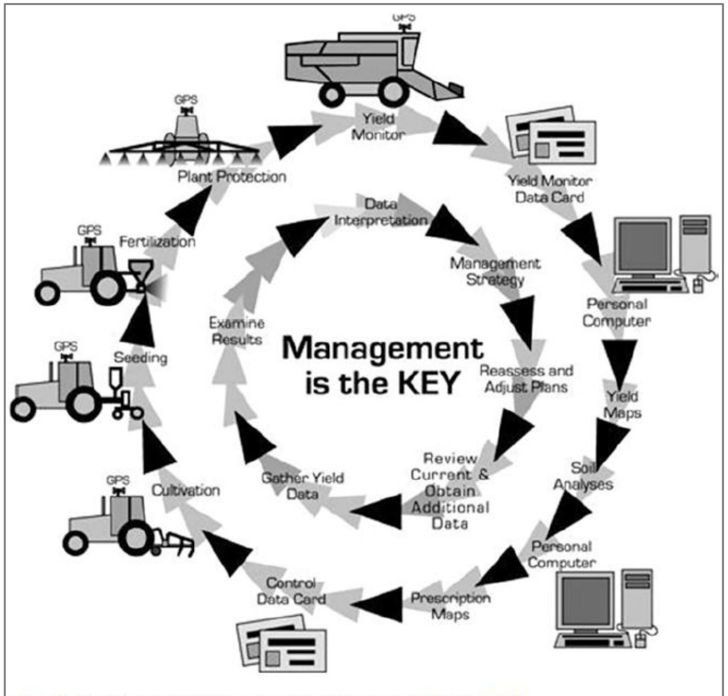


Fig. 1.3 Precision agriculture: a comprehensive approach (Banu 2015)

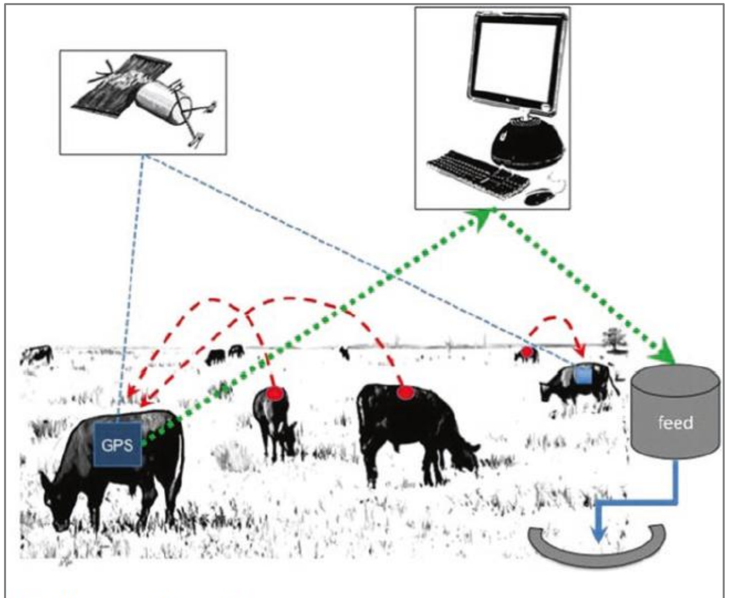


Fig. 3.12 Precision livestock farming

# Components of Precision Agriculture

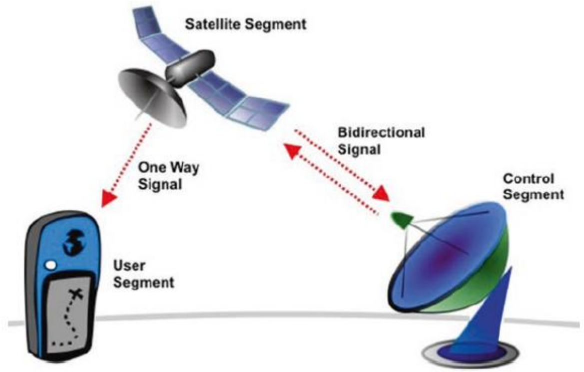



Fig. 3.1 Illustration of the three core segments to a Global Positioning System. (Hinch 2010)

## GPS and Differential Global Positioning System (DGPS)

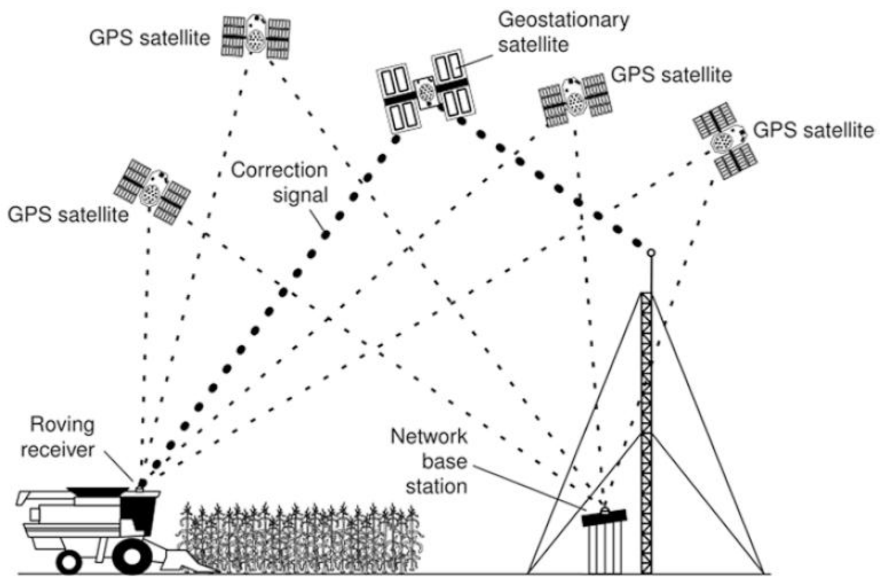



Fig. 2.2 Diagrammatic representation of DGPS

# Components of Precision Agriculture

## Sensor technologies

Fuente:https://aurorasolar.com/blog/how-lidar-is-transforming-remote-solar-system-design/

**Table 3.1** Commonly used sensors in precision agriculture

| Sensor | Applications |
|---|---|
| Temperature sensor | Used to measure temperature with an electrical output proportional to the temperature (in °C). It can measure temperature more accurately than a using a thermostat (Fig. 3.3) |
| Humidity sensor | Used to measure the presence of water in land. The humidity sensor has its output proportional to the temperature (in RH %). The operating range is for humidity is generally 20–95% RH and for temperature is 0–60 °C (Fig. 3.3) |
| See Fig. 3.3 | |
| Water level sensor | Also known as float balls, are spherical, cylindrical, or similarly shaped objects, made from either rigid or flexible material, that are buoyant in water and other liquids. They are non-electrical hardware frequently used as visual sight indicators for surface demarcation and level (Fig. 3.4) |
| Soil sensor | Provide continual, consistent accuracy measuring a number of parameters including the three most significant soil parameters simultaneously – moisture, salinity, and temperature (Fig. 3.5) |
| See Fig. 3.4 | See Fig. 3.5 |
| LIDAR | Light detection and ranging is used to obtain dynamic measurements to estimate fruit-tree leaf area and combined with GPS has been applied for 3-D map generation in vine plantations. Laser and hyperspectral data are used for tree classification, including coniferous and deciduous trees. 3-D modeling of tomato canopies is obtained through high-resolution portable scanning LIDAR (Fig. 3.6) |
| See Fig. 3.6 | |
| Capacitance probes | Suitable for measurement of soil moisture in tropical areas |
| Reflectometers | Can determine moisture content in oil palm fruits |
| Ultrasonic ranging sensors | Used for analyzing apple tree canopies |
| Optoelectronic sensors | Weed detection in wide row crops has been analyzed in terms of accuracy and feasibility |
| Fluorescence-based optical sensor | Used to monitor grape maturation by specifically monitoring anthocyanin accumulation |
| The infrared thermocouple | Measurement of the leaf temperature of forests or agricultural plants (Fig. 3.7) |

Fuente: L. Ahmad, S. Sheraz Mahdi, Satellite farming, An Information and Technology Based Agriculture, Springer Nature Switzerland AG 2018.

# Components of Precision Agriculture

Eddy covariance system consisting of an ultrasonic anemometer and infrared gas analyser.

An eddy correlation instrument measuring oxygen fluxes in benthic environments.

Fuente: https://en.wikipedia.org/wiki/Eddy_covariance

**Table 3.1** (continued)

| Sensor | Applications |
|---|---|
| See Fig. 3.7 | See Fig. 3.8 |
| An FPGA-based fused smart sensor | Real-time plant transpiration monitoring is possible |
| Automated point dendrometers | Analyze tropical tree line stem growth has been applied in forest inventories (Fig. 3.8) |
| Optical and microwave sensors | Suitable for characterizing olive grove canopies |
| Optoelectronic sensors | Weed detection in wide row crops has been analyzed in terms of accuracy and feasibility |
| pH soil-based sensors | Allow measurements of variables in the soil oriented toward crop productivity |
| See (Fig. 3.9) | |
| Eddy covariance sensors | Applied to quantify carbon metabolism of peatlands and also regional and global analysis of observations from micrometeorological tower sites |
| Near-infrared (NIR) sensors | Measure the amount of light reflected back at them at specific wave lengths in order to make this estimate. Estimate the amount of biomass present and calculate future nitrogen requirements |
| | Can be integrated into systems to control fertilizer application at any point in the field |
| | Crop scouting prior to fertilizer application to determine the optimum rate to be applied over the whole field |

# Components of Precision Agriculture



**Fig. 3.3** Humidity and temperature sensor on a California farm

**Fig. 3.5** Soil sensor

# Components of Precision Agriculture



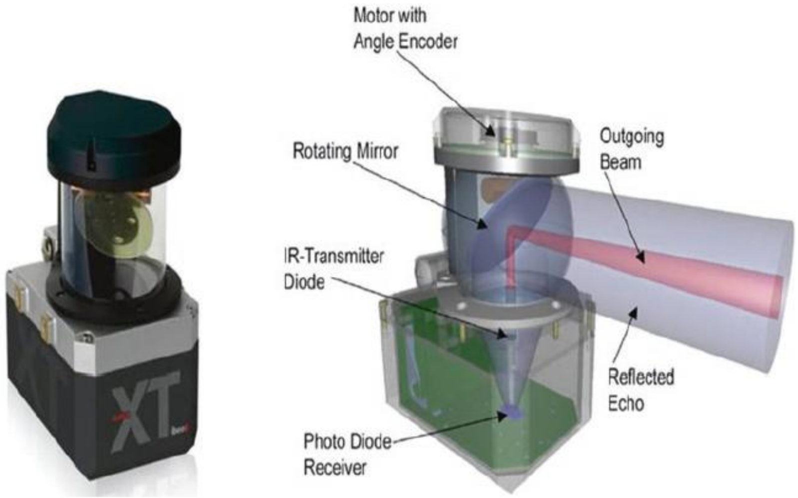
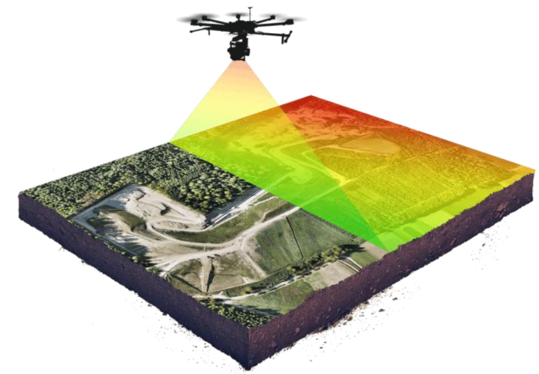


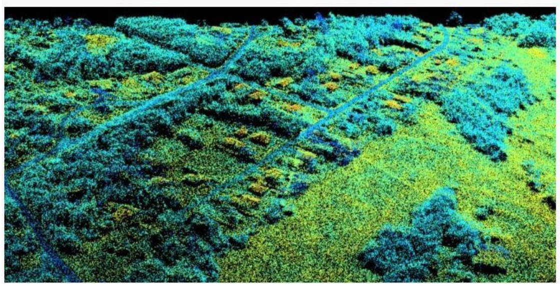

Fig. 3.6 LIDAR

Fig. 3.7 Infrared thermocouple

[1]

## LiDAR (Light Detection And Ranging)





[2]

Fuente: [1] L. Ahmad, S. Sheraz Mahdi, Satellite farming, An Information and Technology Based Agriculture, Springer Nature Switzerland AG 2018.
[2] Fuente: https://wingtra.com/es/dron-fotogrametria-vs-lidar/

# Components of Precision Agriculture

Fig. 3.8 Automated point dendrometer

Geographic information system (GIS)

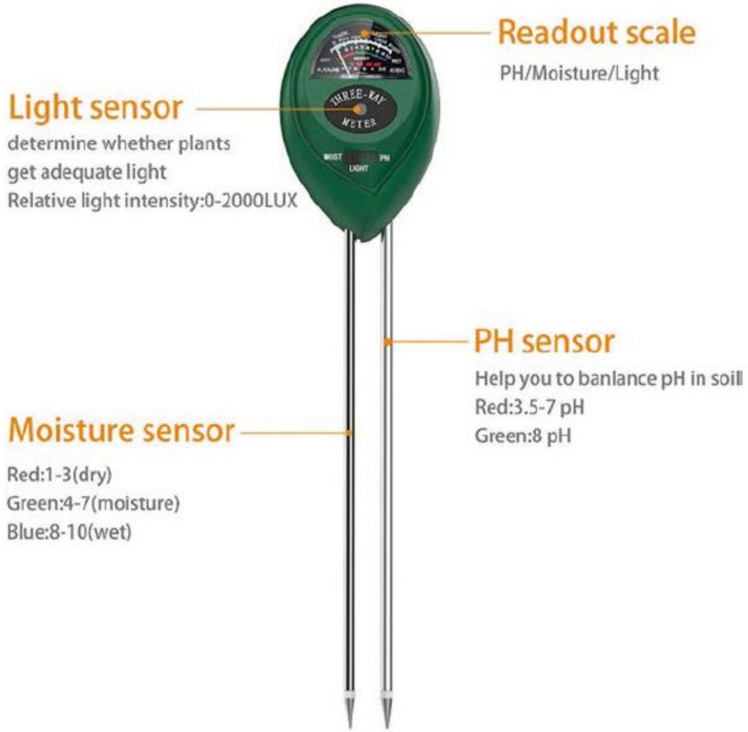



Fig. 3.9 pH meter with sensor

Grid Soil Sampling and Variable Rate Fertilizer Application

# Components of Precision Agriculture

Precision Farming Within the Fruits and Vegetables and Viticulture Sectors

Precision Farming on Arable Land

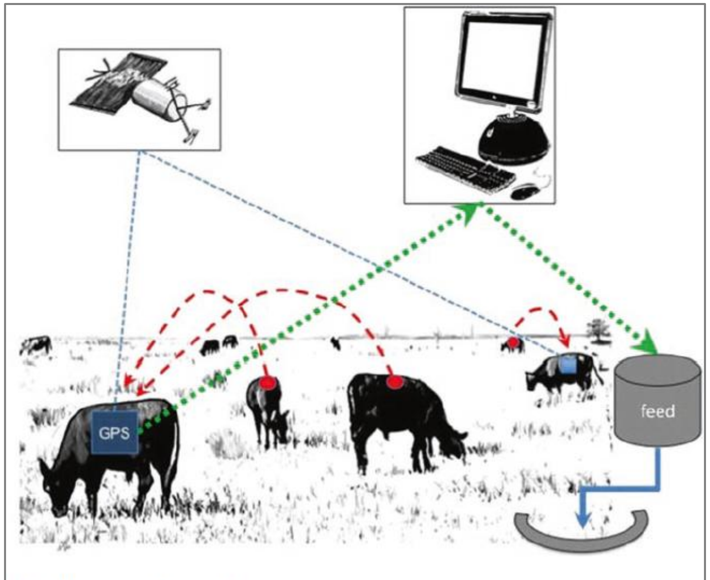



Fig. 3.12 Precision livestock farming

Precision Livestock Farming

Rate Controllers

Fig. 3.10 Display monitor of a rate controller

# Components of Precision Agriculture

Variable Rate Controllers / Applications

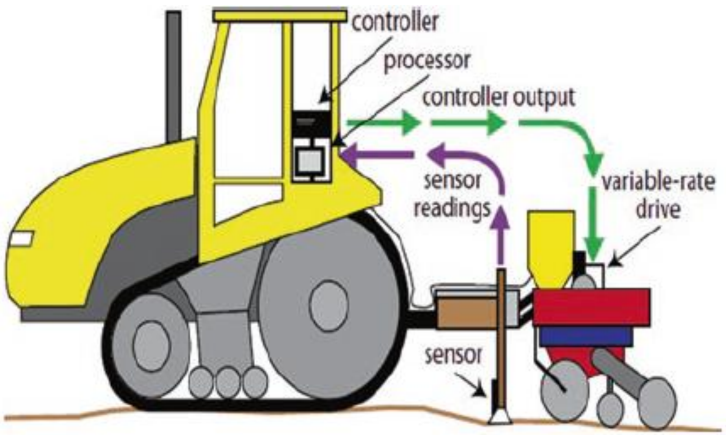



**Fig. 5.1** On the go sensor measuring the soil characteristics before planting and adjusting seed rate. (Grisso et al. 2011)

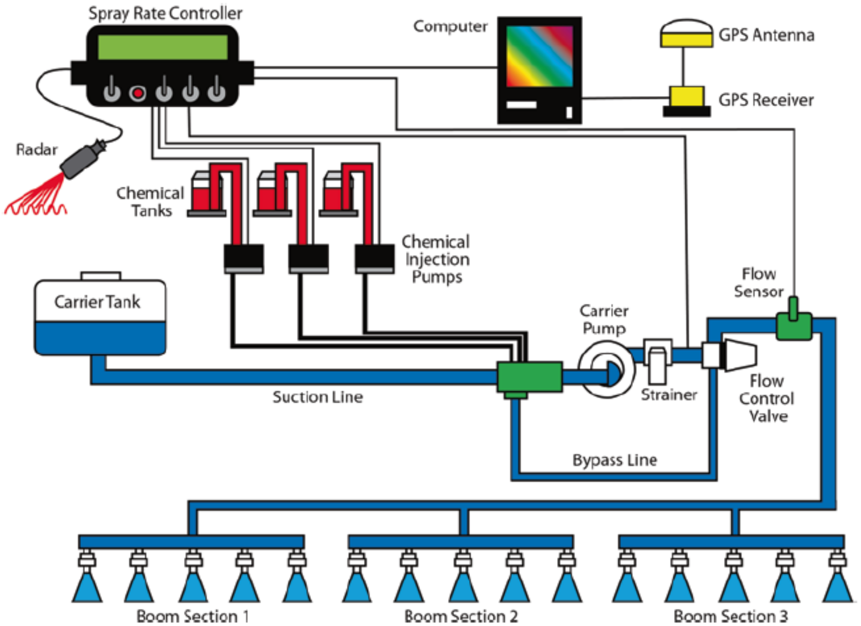



**Fig. 5.3** VRA spraying system that involves chemical injection technology. (Grisso et al. 2011)

Fuente: L. Ahmad, S. Sheraz Mahdi, Satellite farming, An Information and Technology Based Agriculture, Springer Nature Switzerland AG 2018.

# Components of Precision Agriculture

Variable Rate Controllers / Applications

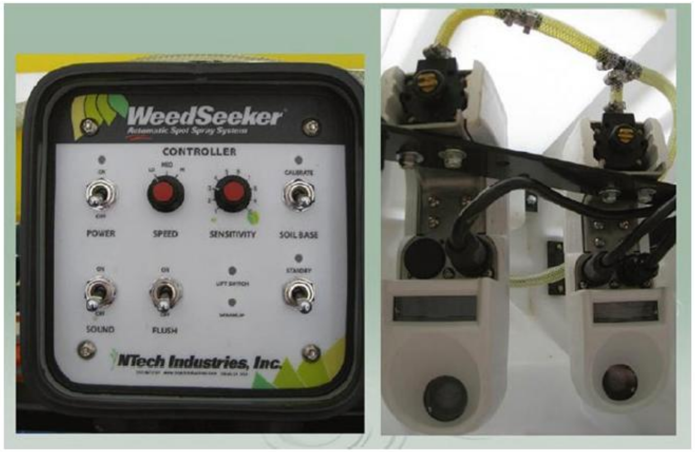
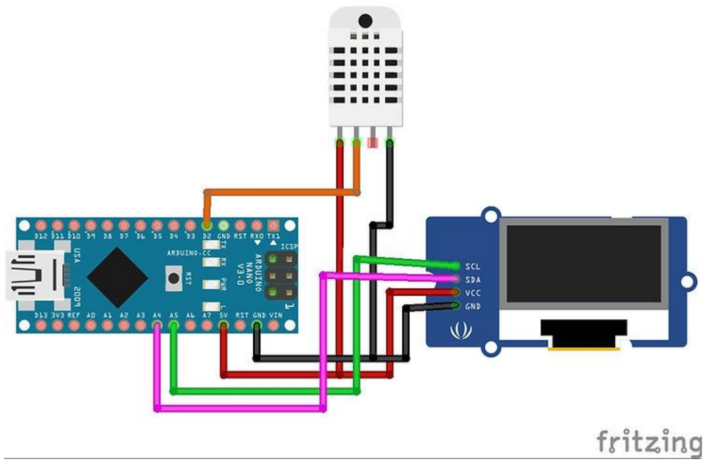


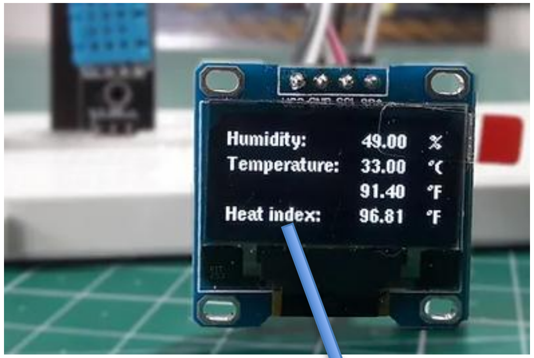

Fig. 5.6 Weed seeker, controller, and sensors

Fig. 5.7 Weed seeker-efficient cost-effective and environmentally sustainable

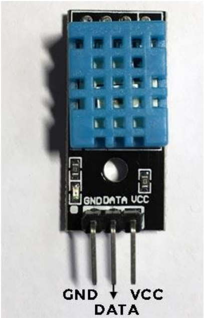
# Electrónica para agricultura

## Arduino | Medición de humedad relativa y temperatura en aire con DTH 11



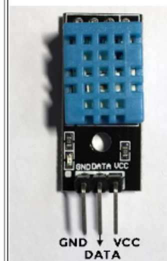

El DHT11 es un sensor de temperatura y humedad que utiliza un sensor capacitivo de humedad y un termistor para medir el aire circundante, y muestra los datos mediante una señal digital en el pin de datos




Heat index: parámetro que tiene en cuenta la temperatura y la humedad relativa, determina la temperatura aparente o percibida por el ser humano.

# Electrónica para agricultura

## Arduino | Medición de humedad relativa y temperatura en aire con DTH 11

| Características | | |
|---|---|---|
| Rango de medición humedad: 20% a 90% RH<br>Rango de medición temperatura: 0ºC a +50°C<br>Resolución humedad: +/- 5.0% RH<br>Resolución temperatura: +/- 2.0ºC<br>Tamaño compacto: 2.3cm x 1.2cm x 0.5cm.<br>Alimentación de 3.3V a 5VDC<br>Corriente máxima 2.5mA durante la conversión.<br>No más de 1 Hz en velocidad de muestreo (una vez cada segundo) |  | **Pines de la versión PCB del DHT11**<br>**GND: Conexión con tierra**<br>**DATA: Transmisión de datos**<br>**VCC: Alimentación** |

[1]

The heat index formula is expressed as,

$$HI= c_1+c_2T+c_3R+c_4TR+c_5T_2+c_6R_2+c_7T_2R+c_8TR_2+c_9T_2R_2$$

In this formula,
HI = heat index in degrees Fahrenheit
R = Relative humidity
T = Temperature in ∘F
$c_1 = -42.379$
$c_2 = -2.04901523$
$c_3 = -10.14333127$
$c_4 = -0.22475541$
$c_5 = -6.83783 \times 10^{-3}$
$c_6 = -5.481717 \times 10^{-2}$
$c_7 = -1.22874 \times 10^{-3}$
$c_8 = 8.5282 \times 10^{-4}$
$c_9 = -1.99 \times 10^{-6}$

[2]

**NOAA national weather service: heat index**

| Temperature / Relative humidity | 80 °F (27 °C) | 82 °F (28 °C) | 84 °F (29 °C) | 86 °F (30 °C) | 88 °F (31 °C) | 90 °F (32 °C) | 92 °F (33 °C) | 94 °F (34 °C) | 96 °F (36 °C) | 98 °F (37 °C) | 100 °F (38 °C) | 102 °F (39 °C) | 104 °F (40 °C) | 106 °F (41 °C) | 108 °F (42 °C) | 110 °F (43 °C) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **40%** | 80 °F (27 °C) | 81 °F (27 °C) | 83 °F (28 °C) | 85 °F (29 °C) | 88 °F (31 °C) | 91 °F (33 °C) | 94 °F (34 °C) | 97 °F (36 °C) | 101 °F (38 °C) | 105 °F (41 °C) | 109 °F (43 °C) | 114 °F (46 °C) | 119 °F (48 °C) | 124 °F (51 °C) | 130 °F (54 °C) | 136 °F (58 °C) |
| **45%** | 80 °F (27 °C) | 82 °F (28 °C) | 84 °F (29 °C) | 87 °F (31 °C) | 89 °F (32 °C) | 93 °F (34 °C) | 96 °F (36 °C) | 100 °F (38 °C) | 104 °F (40 °C) | 109 °F (43 °C) | 114 °F (46 °C) | 119 °F (48 °C) | 124 °F (51 °C) | 130 °F (54 °C) | 137 °F (58 °C) | |
| **50%** | 81 °F (27 °C) | 83 °F (28 °C) | 85 °F (29 °C) | 88 °F (31 °C) | 91 °F (33 °C) | 95 °F (35 °C) | 99 °F (37 °C) | 103 °F (39 °C) | 108 °F (42 °C) | 113 °F (45 °C) | 118 °F (48 °C) | 124 °F (51 °C) | 131 °F (55 °C) | 137 °F (58 °C) | | |
| **55%** | 81 °F (27 °C) | 84 °F (29 °C) | 86 °F (30 °C) | 89 °F (32 °C) | 93 °F (34 °C) | 97 °F (36 °C) | 101 °F (38 °C) | 106 °F (41 °C) | 112 °F (44 °C) | 117 °F (47 °C) | 124 °F (51 °C) | 130 °F (54 °C) | 137 °F (58 °C) | | | |
| **60%** | 82 °F (28 °C) | 84 °F (29 °C) | 88 °F (31 °C) | 91 °F (33 °C) | 95 °F (35 °C) | 100 °F (38 °C) | 105 °F (41 °C) | 110 °F (43 °C) | 116 °F (47 °C) | 123 °F (51 °C) | 129 °F (54 °C) | 137 °F (58 °C) | | | | |
| **65%** | 82 °F (28 °C) | 85 °F (29 °C) | 89 °F (32 °C) | 93 °F (34 °C) | 98 °F (37 °C) | 103 °F (39 °C) | 108 °F (42 °C) | 114 °F (46 °C) | 121 °F (49 °C) | 128 °F (53 °C) | 136 °F (58 °C) | | | | | |
| **70%** | 83 °F (28 °C) | 86 °F (30 °C) | 90 °F (32 °C) | 95 °F (35 °C) | 100 °F (38 °C) | 105 °F (41 °C) | 112 °F (44 °C) | 119 °F (48 °C) | 126 °F (52 °C) | 134 °F (57 °C) | | | | | | |
| **75%** | 84 °F (29 °C) | 88 °F (31 °C) | 92 °F (33 °C) | 97 °F (36 °C) | 103 °F (39 °C) | 109 °F (43 °C) | 116 °F (47 °C) | 124 °F (51 °C) | 132 °F (56 °C) | | | | | | | |
| **80%** | 84 °F (29 °C) | 89 °F (32 °C) | 94 °F (34 °C) | 100 °F (38 °C) | 106 °F (41 °C) | 113 °F (45 °C) | 121 °F (49 °C) | 129 °F (54 °C) | | | | | | | | |
| **85%** | 85 °F (29 °C) | 90 °F (32 °C) | 96 °F (36 °C) | 102 °F (39 °C) | 110 °F (43 °C) | 117 °F (47 °C) | 126 °F (52 °C) | 135 °F (57 °C) | | | | | | | | |
| **90%** | 86 °F (30 °C) | 91 °F (33 °C) | 98 °F (37 °C) | 105 °F (41 °C) | 113 °F (45 °C) | 122 °F (50 °C) | 131 °F (55 °C) | | | | | | | | | |
| **95%** | 86 °F (30 °C) | 93 °F (34 °C) | 100 °F (38 °C) | 108 °F (42 °C) | 117 °F (47 °C) | 127 °F (53 °C) | | | | | | | | | | |
| **100%** | 87 °F (31 °C) | 95 °F (35 °C) | 103 °F (39 °C) | 112 °F (44 °C) | 121 °F (49 °C) | 132 °F (56 °C) | | | | | | | | | | |

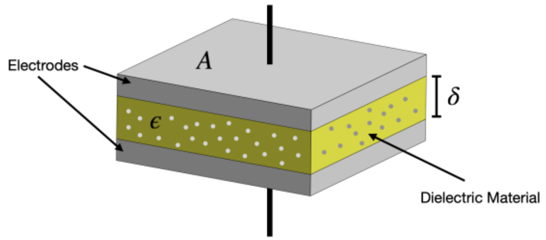
[3]

*Key to colors:* Caution | Extreme caution | Danger | Extreme danger

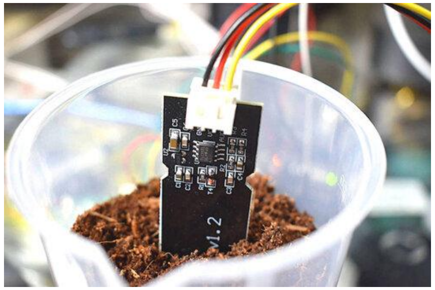
Fuente: [1] http://ceca.uaeh.edu.mx/informatica/oas_final/OA4/temperatura_y_humedad_dht11.html
[2] https://byjus.com/heat-index-formula/
[3] Wikipedia contributors, "Heat index," Wikipedia, The Free Encyclopedia, https://en.wikipedia.org/w/index.php?title=Heat_index&oldid=1171373578 (accessed October 2[illegible]

# Electrónica para agricultura

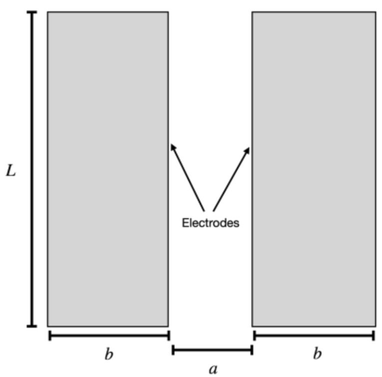
## Arduino | Medición de humedad en suelos (*Capacitive Soil Moisture sensor*)

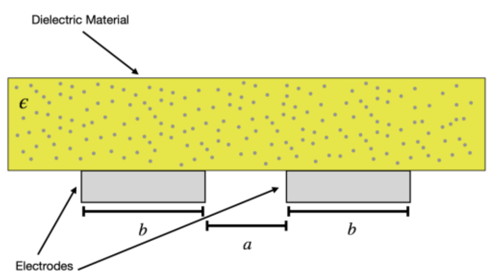


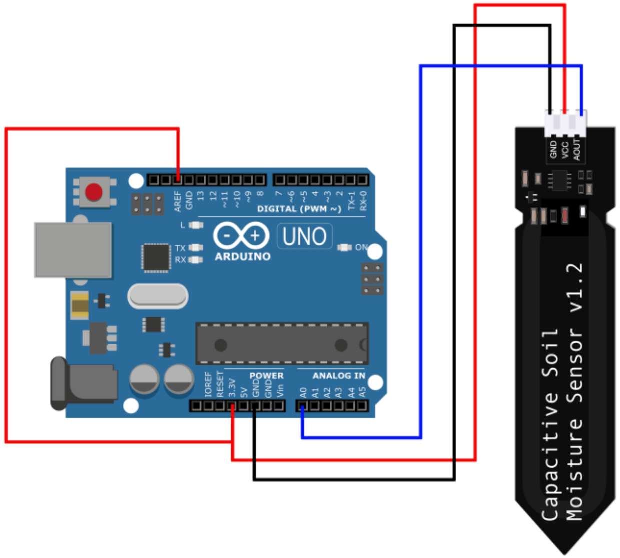

$$C = \frac{\epsilon \mathbf{E} A}{\mathbf{E}\delta} = \frac{\epsilon A}{\delta}$$




Las placas son planas y están una al lado de la otra, en lugar de una encima de la otra. El material dieléctrico es el suelo.

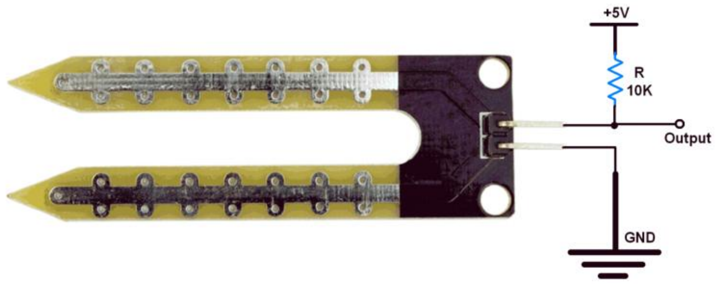


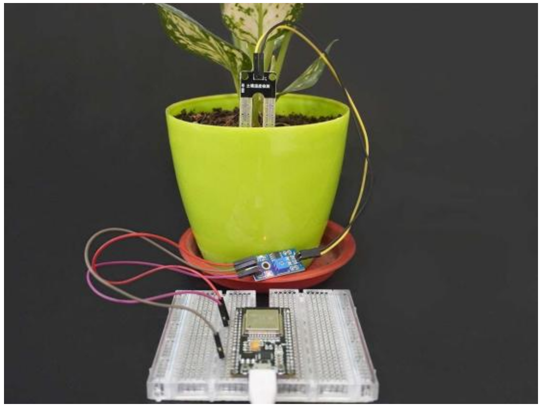



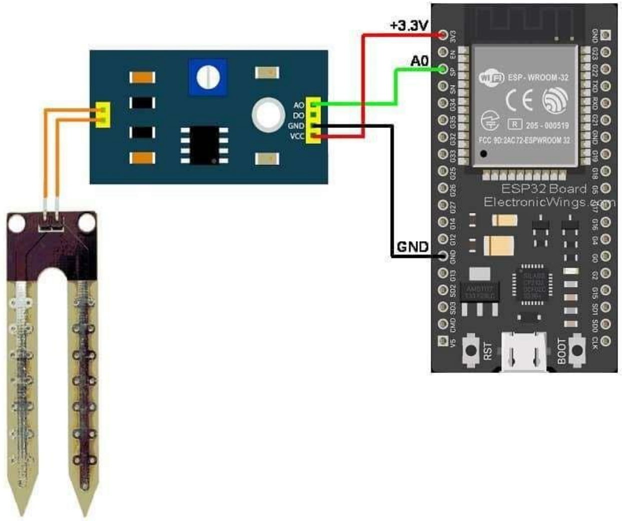



# Electrónica para agricultura

ESP32 | Medición de humedad en suelos (*Resistive Soil Moisture sensor*)

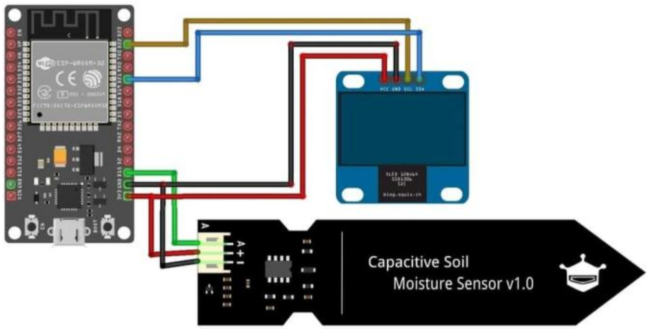


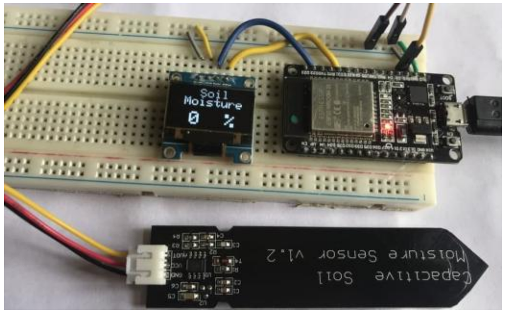



Las tarjetas ESPXX facilitan la conexión con sensores, periféricos, la comunicación con dispositivos móviles, y puede programarse con el Arduino IDE

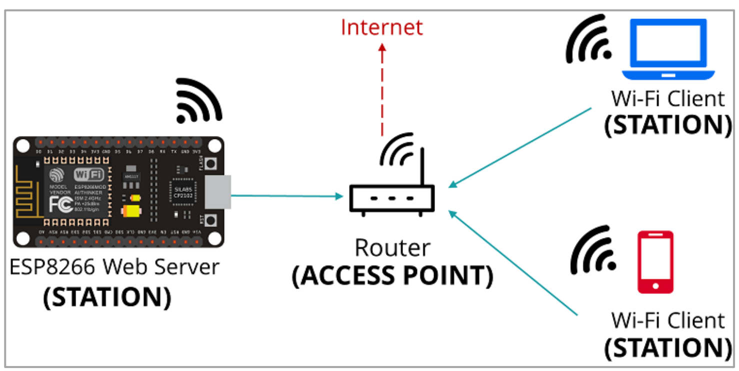
## ESP32 | Medición de humedad en suelos (*Capacitive Soil Moisture sensor*)

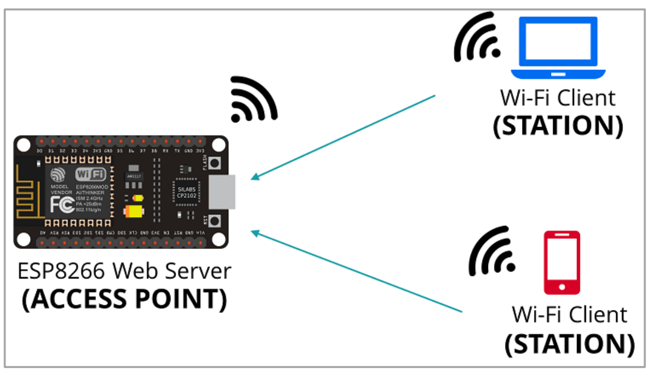



[1]

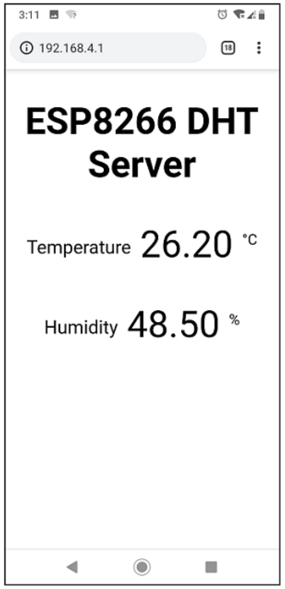



Las tarjetas ESPXX facilitan la conexión inalámbrica por WiFi o Bluetooth

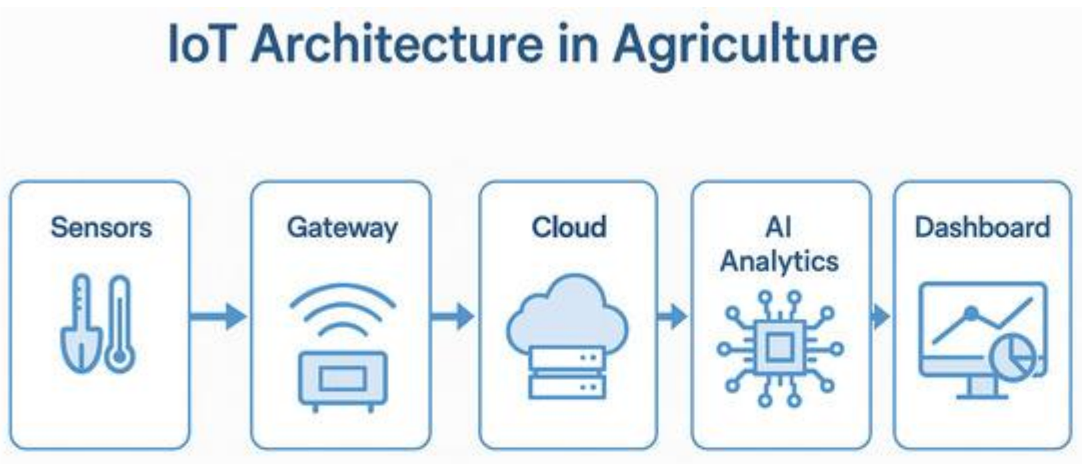


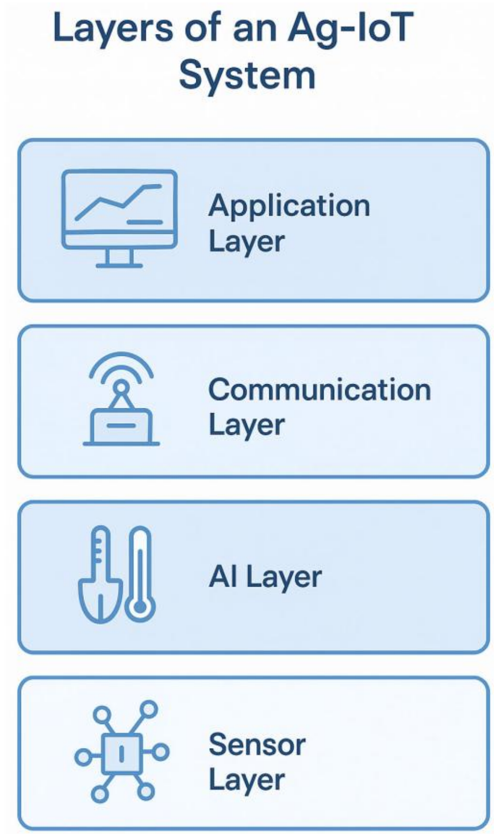




[2]



# IoT para agricultura




**The IoT and AI in Agriculture: The Time Is Now—A Systematic Review of Smart Sensing Technologies**

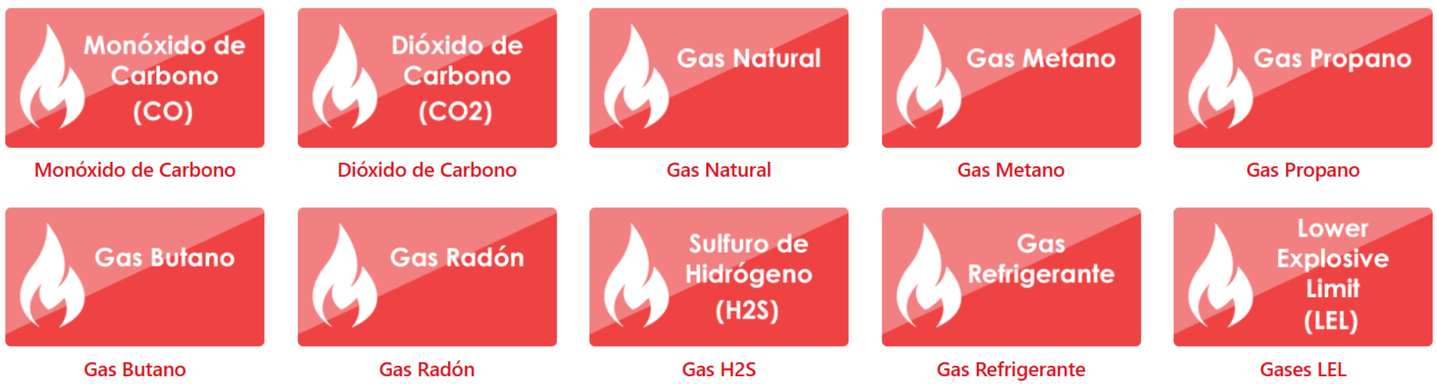
by Tymoteusz Miller [1,2,*], Grzegorz Mikiciuk [3], Irmina Durlik [4], Małgorzata Mikiciuk [5], Adrianna Łobodzińska [6,7] and Marek Śnieg [8]

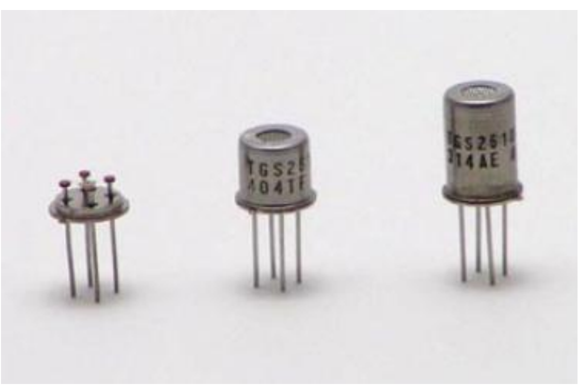
Fuente: [1] https://how2electronics.com/capacitive-soil-moisture-sensor-esp8266-esp32-oled-display/
[2] https://randomnerdtutorials.com/esp8266-nodemcu-access-point-ap-web-server/

## Sensores de Gas

Tienen gran aplicación en la monitorización de la calidad del aire en interiores, hasta la creación de detectores de gas personalizados para entornos domésticos y comerciales

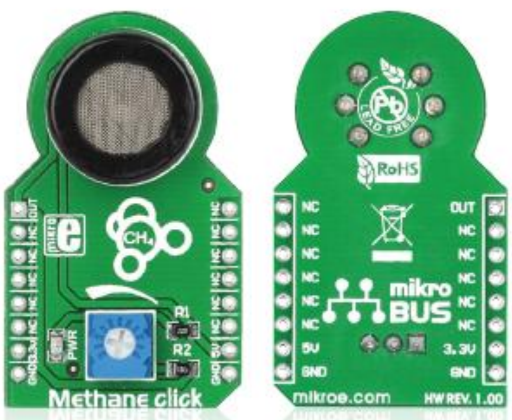
Detectores de Gas por Tipo de Gas

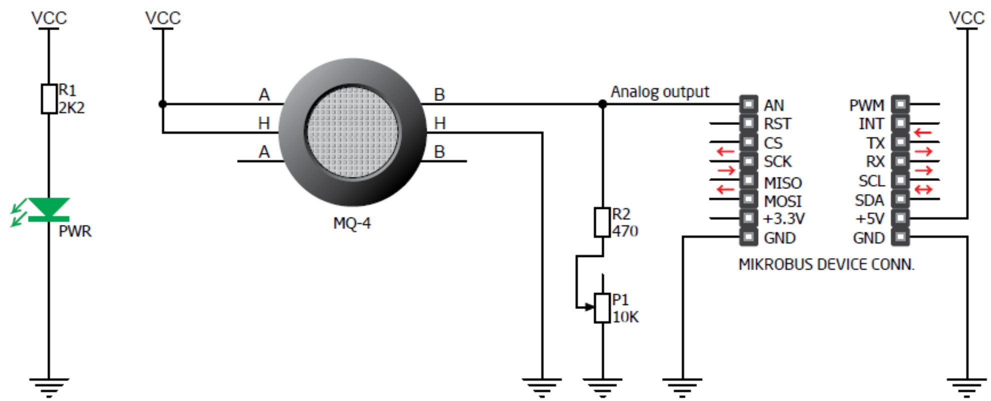



Fuente: https://detectoresgas.com/formato/sensores-mq/

Methane click™ is a simple solution for adding a high sensitivity methane ($CH_4$) sensor to your design. It's suitable for designing gas leakage equipment.




# Electrónica para agricultura

## Sensores de Gas

HANWEI ELETRONICS CO.,LTD MQ-2 http://www.hwsensor.com

# TECHNICAL DATA MQ-2 GAS SENSOR

### FEATURES

Wide detecting scope Fast response and High sensitivity
Stable and long life Simple drive circuit

### APPLICATION

They are used in gas leakage detecting equipments in family and industry, are suitable for detecting of LPG, i-butane, propane, methane ,alcohol, Hydrogen, smoke.

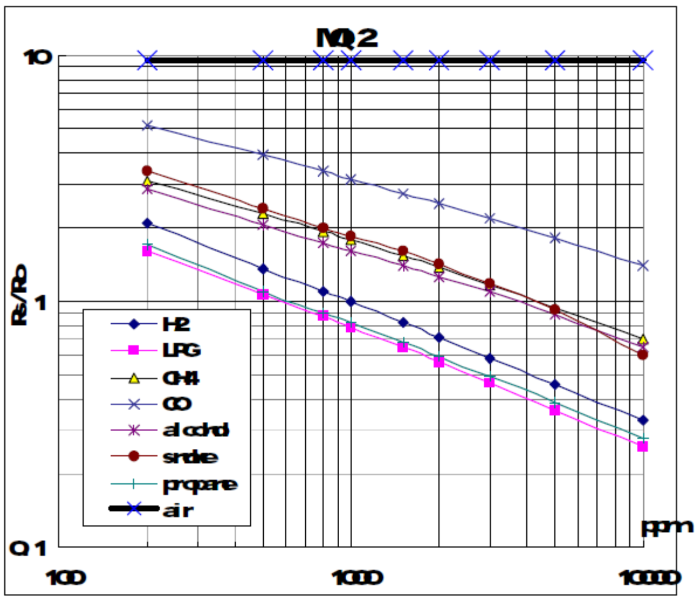



Fig.2 sensitivity characteristics of the MQ-2

| | Parts | Materials |
|---|---|---|
| 1 | Gas sensing layer | $SnO_2$ |
| 2 | Electrode | Au |
| 3 | Electrode line | Pt |
| 4 | Heater coil | Ni-Cr alloy |
| 5 | Tubular ceramic | $Al_2O_3$ |
| 6 | Anti-explosion network | Stainless steel gauze (SUS316 100-mesh) |
| 7 | Clamp ring | Copper plating Ni |
| 8 | Resin base | Bakelite |
| 9 | Tube Pin | Copper plating Ni |

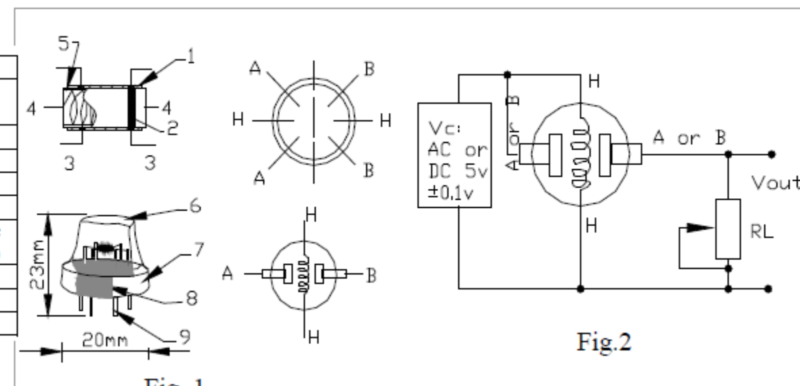



Fig. 1

Fig.2


LoRa Communication for Agriculture 4.0:
Opportunities, Challenges, and Future Directions

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Fuente: https://detectoresgas.com/formato/sensores-mq/